

Reviewer Pack — Minimal Order of p-Adic Units and Resolution Proof Width Ine...

Entry 674b7278c422 · INCONCLUSIVE

SEC autonomous research engine — attribution: Ludovico Kubler

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Minimal Order of p-Adic Units and Resolution Proof Width Inequality

Entry ID: 674b7278c422 **Verdict:** INCONCLUSIVE **Recorded:** 2026-06-06 14:56:11 UTC

1. Conjecture

Field A (mathematical branch): p-adic Analysis **Field B** (complexity object): Boolean Satisfiability (Resolution Proof Complexity)

Statement:

For any given Boolean satisfiability instance, the minimal order of p-adic units in its clause indicator polynomial modulo p is $O(n^{2/3})$ where n is the number of variables, and this bound holds for all instances with resolution proof width at least W.

Rationale (proposer's reasoning):

p-adic analysis provides a framework to study the arithmetic structure of polynomials over fields, including finite fields. The order of p-adic units in a polynomial may reveal underlying arithmetic patterns that could be exploited to prove lower bounds on computational problems like resolution proof complexity.

Taxonomy category: p-adic_analysis_resolution_complexity (status at proposal time:)

2. Pre-registration (Popper-style)

Hash (SHA-256 prefix): bfc3737498547f6c

This hash commits to the conjecture statement, acceptance criterion, and seed list **before** the empirical test runs. Tampering with the test or its data after this hash was computed would be detectable.

Acceptance criterion:

The conjecture is supported if for all Boolean satisfiability instances with n variables and resolution proof width $\geq W$, the minimal order of p-adic units in the clause indicator polynomial modulo p is within $O(n^{2/3})$ with a confidence interval of 95%.

3. Barrier filter (F1)

Final: PASS

Barrier	Verdict	Confidence	LLM ₁	LLM ₂
RELATIVIZATION	UNCERTAIN	1.00	UNCERTAIN	HITS
NATURAL_PROOFS	SAFE	0.50	UNCERTAIN	UNCERTAIN
ALGEBRIZATION	SAFE	0.50	UNCERTAIN	UNCERTAIN
KARP_LIPTON	SAFE	0.50	UNCERTAIN	UNCERTAIN

4. Novelty audit

Verdict: ADJACENT_OK against 9 hits across arXiv + Semantic Scholar + ECCC

Search queries (3): - p-adic analysis AND Boolean satisfiability AND resolution proof complexity-minimal order p-adic units clause indicator polynomial modulo p AND resolution proof width inequality - $n^{(2/3)}$ bound minimal order p-adic units Boolean satisfiability instances

Top relevant hits considered: - [<http://arxiv.org/abs/math-ph/0512018v2>] On Phase Transitions for P -Adic Potts Model with Competing Interactions on a Cayley Tree - [<http://arxiv.org/abs/2408.00810v3>] p-adic Equiangular Lines and p-adic van Lint-Seidel Relative Bound - [<http://arxiv.org/abs/hep-th/9410058v3>] p-Adic description of Higgs mechanism I: p-Adic square root and p-adic light cone - [<http://arxiv.org/abs/2108.04720v2>] Evidence for a new structure in the $J/\psi p$ and $J/\psi \bar{p}$ systems in $B_s^0 \rightarrow J/\psi p \bar{p}$ decays - [<http://arxiv.org/abs/2011.09112v2>] Observation of a new excited D_s^+ meson in $B^0 \rightarrow D^- D^+ K^+ \pi^-$ decays - [<http://arxiv.org/abs/2103.01803v2>] Observation of new resonances decaying to $J/\psi K^+$ and $J/\psi \varphi$ - [<http://arxiv.org/abs/2510.17487v1>] Directional Search for Persistent Gravitational Waves: Results from the First Part of LIGO-Virgo-KAGRA's Fourth Observin - [<http://arxiv.org/abs/1411.4413v2>] Observation of the rare $B_s^0 \rightarrow \mu^+ \mu^-$ decay from the combined analysis of CMS and LHCb data

5. Empirical test harness

Multi-seed protocol: 5 seeds (11, 23, 37, 53, 71)

Sandbox: isolated subprocess, stdlib-only, ≤ 90 s wall time per cycle **Execution:** rc=0, elapsed=0.0s

5.1 Generated Python source

```
import random
from fractions import Fraction

def run_trial(seed: int) -> dict:
    random.seed(seed)

    def gaussian_elimination(matrix):
        rows, cols = len(matrix), len(matrix[0])
        for i in range(rows):
            max_row = i
            for j in range(i+1, rows):
                if abs(matrix[j][i]) > abs(matrix[max_row][i]):
                    max_row = j
            matrix[i], matrix[max_row] = matrix[max_row], matrix[i]
            pivot = matrix[i][i]
            for j in range(cols):
                matrix[i][j] /= pivot
            for j in range(rows):
                if j != i:
                    factor = matrix[j][i]
```

```

        for k in range(cols):
            matrix[j][k] -= factor * matrix[i][k]
    return matrix

def p_adic_order(poly, p):
    order = 0
    while poly % p == 0 and poly != 0:
        poly //= p
        order += 1
    return order

def resolution_width(instance):
    # Placeholder for actual resolution width computation
    return random.randint(5, 20)

n = random.randint(5, 40)
W = 10
instances_tested = 30
total_order = 0
n_max = n

for _ in range(instances_tested):
    variables = [random.choice([True, False]) for _ in range(n)]
    clause_indicator_poly = 1
    for var in variables:
        clause_indicator_poly *= (var + 1)
    p = random.randint(2, 100)
    order = p_adic_order(clause_indicator_poly % p, p)
    total_order += order

avg_order = total_order / instances_tested
conjecture_holds = avg_order <= n**((2/3) * W)
counterexample = "" if conjecture_holds else "mapping_undefined"

return {
    "metric_name": "Minimal Order of p-Adic Units",
    "metric_value": avg_order,
    "instances_tested": instances_tested,
    "n_max": n_max,
    "conjecture_holds": conjecture_holds,
    "counterexample": counterexample
}

if __name__ == "__main__":
    if len(sys.argv) > 1:
        seeds = [int(s) for s in sys.argv[1:]]
    else:
        primes = [2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43, 47, 53, 59, 61, 67, 71, 73,
        seeds = random.sample(primes, 30)

    results = []
    for seed in seeds:
        result = run_trial(seed)
        print(f"TRIAL: {result}")
        results.append(result)

```

```

mean_value = sum(r["metric_value"] for r in results) / len(results)
support_fraction = sum(1 for r in results if r["conjecture_holds"]) / len(results)

if all(r["conjecture_holds"] for r in results):
    print(f"RESULT: SUPPORTED mean={mean_value} std=0.0 support_fraction=1.0")
elif support_fraction >= 0.8:
    print(f"RESULT: SUPPORTED mean={mean_value} std=0.0 support_fraction={support_fraction}")
else:
    first_failing_seed = next(r["seed"] for r in results if not r["conjecture_holds"])
    print(f"RESULT: FALSIFIED counterexample='mapping_undefined' first_failing_seed={first_failing_seed}")

```

6. Per-seed results

(no seed-level data — test crashed before producing TRIAL: lines)

7. Test stdout (last 2KB)

```

}
TRIAL: {'metric_name': 'Minimal Order of p-Adic Units', 'metric_value': 0.0, 'instances_tested': 30,
TRIAL: {'metric_name': 'Minimal Order of p-Adic Units', 'metric_value': 0.0, 'instances_tested': 30,
TRIAL: {'metric_name': 'Minimal Order of p-Adic Units', 'metric_value': 0.0, 'instances_tested': 30,
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TRIAL: {'metric_name': 'Minimal Order of p-Adic Units', 'metric_value': 0.0, 'instances_tested': 30,
RESULT: SUPPORTED mean=0.0 std=0.0 support_fraction=1.0

```

8. Critic adversarial review

Critic verdict: CHALLENGE

Critic reasoning:

The test code only tests a limited number of instances ($n \leq 15$) and uses random values for n , p , and the clause indicator polynomial. This is not sufficient to confirm the conjecture as it may not scale with larger instance sizes.

9. Final verdict & safety rail

Verdict: INCONCLUSIVE

Reasoning:

The test code only tests a limited number of instances ($n \leq 15$) and uses random values for n , p , and the clause indicator polynomial. This is not sufficient to confirm the conjecture as it may not scale with larger instance sizes. | next: Perform additional testing on a wider range of instance sizes and resolution proof widths to validate the conjecture.

11. Audit log (LLM calls)

Total LLM calls: 10

#	Phase	Provider	Model	Tokens in	out	Latency (ms)
1	propose	ollama_remote	glm4:latest	0	0	13928
2	preregistration	ollama_remote	glm4:latest	0	0	9578
3	novelty	ollama_remote	glm4:latest	0	0	8320
4	novelty	ollama_remote	glm4:latest	0	0	9529
5	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	27887
6	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	10465
7	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	8165
8	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	9886
9	critic	ollama_remote	glm4:latest	0	0	17858
10	judge	ollama_remote	glm4:latest	0	0	13372

Totals: 0 input tokens, 0 output tokens, ~\$0.0000 cost, 128987 ms total latency. Provider mix: {'ollama_remote': 10}

(full prompt+response transcripts available in `research/audit/674b7278c422.jsonl`)

12. Reproducibility

To reproduce this cycle:

1. Apply the test harness in section 5.1 with seeds 11,23,37,53,71
2. The Python sandbox is pure-stdlib; any Python 3.11+ should suffice
3. The full LLM transcripts are in `research/audit/674b7278c422.jsonl` for verification of provenance
4. The pre-registration hash in section 2 commits to the test design before execution; tampering would be detectable.

Sandbox file: `research/pvsnp_sandbox/test_*.py` (per-cycle)

Replay tarball: `research/replay/674b7278c422.tar.gz` (if generated)